

AZITHRO
(Azithromycin)

20180913AA

1. COMPOSITION

1.1 Azithro (Capsule)

Each capsule contains: Azithromycin dihydrate equivalent to azithromycin 250 mg

2. PHARMACOLOGICAL PROPERTIES

Azithromycin is the first of a subclass of macrolide antibiotics, known as azalides. The mode of action of azithromycin is inhibition of protein synthesis in bacteria by binding to the 50s ribosomal subunit and preventing translocation of peptides.

Azithromycin demonstrates activity in vitro against a wide range of bacteria including:

Gram-positive Aerobic Bacteria - *Staphylococcus aureus*, *Streptococcus agalactiae*, *Streptococcus pneumoniae*, *Streptococci* (Group C,F,G), alpha-haemolytic streptococci (viridans group) and other streptococci, and *Corynebacterium diphtheriae*. Azithromycin demonstrates cross-resistance with erythromycin-resistant Gram-positive strains, including *Streptococcus faecalis* (enterococcus) and most strains of methicillin-resistant staphylococci. Macrolide-resistant *Streptococcus pyogenes* (group A beta-hemolytic streptococci).

Gram-negative Aerobic Bacteria - *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Moraxella catarrhalis*, *Acinetobacter* species, *Yersinia* species, *Legionella pneumophila*, *Bordetella pertussis*, *Bordetella parapertussis*, *Shigella* species, *Pasteurella* species, *Vibrio cholerae* and *parahaemolyticus*, *Plesiomonas shigelloides*. Activities against *Escherichia coli*, *Salmonella enteritidis*, *Salmonella typhi*, *Enterobacter* species, *Aeromonas hydrophila* and *Klebsiella* species are variable and susceptibility tests should be performed. *Proteus* species, *Serratia* species, *Morganella* species and *Pseudomonas aeruginosa* are usually resistant.

Anaerobic Bacteria - *Bacteroides fragilis* and *Bacteroides* species, *Clostridium perfringens*, *Peptococcus* species and *Peptostreptococcus* species, *Fusobacterium necrophorum* and *Propionibacterium acnes*.

Organisms of Sexually Transmitted Diseases - Azithromycin is active against *Chlamydia trachomatis* and also shows good activity against *Treponema pallidum*, *Neisseria gonorrhoeae*, and *Haemophilus ducreyi*.

Other Organisms - *Borrelia burgdorferi* (Lyme disease agent), *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, *Mycoplasma hominis*, *Ureaplasma urealyticum*, *Campylobacter* species and *Listeria monocytogenes*.

Opportunistic Pathogens Associated with HIV Infections - *Mycobacterium avium-intracellular* complex, *Pneumocystis carinii* and *Toxoplasma gondii*.

Inherently resistant organisms - *Enterobacteriaceae*, *Pseudomonas*.

3. CLINICAL PROPERTIES

3.1 Pharmacokinetics

The pharmacokinetic profile of azithromycin is characterized by low plasma concentrations but high and persistent tissue concentrations.

Absorption : Bioavailability after oral administration is approximately 37%. Peak plasma concentrations are attained 2-3 hours after taking the medicinal product.
Distribution : Orally administered azithromycin is widely distributed throughout the body. In pharmacokinetic studies it has been demonstrated that the concentrations of azithromycin measured in tissues are noticeably higher (as much as 50 times) than those measured in plasma, which indicates that the agent strongly binds to tissues. Binding to serum proteins varies according to plasma concentration and ranges from 12% at 0.5 mcg/ml up to 52% at 0.05 mcg azithromycin/ml serum. The mean volume of distribution at steady state (Vss) has been calculated to be 31.1 L/kg.

Elimination : The terminal plasma elimination half-life closely reflects the elimination half-life from tissues of 2-4 days. Approximately 12% of an intravenously administered dose of azithromycin is excreted unchanged in urine within the following three days. Particularly high concentrations of unchangedazithromycin have been found in human bile. Also in bile, ten metabolites were detected, which were formed through N-and O-demethylation, hydroxylation of desosamine and aglycone rings and cleavage of cladinose conjugate. Comparison of the results of liquid chromatography and microbiological analyses has shown that the metabolites of azithromycin are not microbiologically active.

In animal tests, high concentrations of azithromycin have been found in phagocytes. It has also been established that during active phagocytosis, higher concentrations of azithromycin are released from inactive phagocytes. In animal models, this results in high concentrations of azithromycin being delivered to the site of infection.

3.2 Therapeutic Indications

Azithro is indicated for infections caused by susceptible organisms: in lower respiratory tract infections including bronchitis and pneumonia, in odontostomatological infections in skin and soft tissue infections. In acute otitis media and in upper respiratory tract infections including sinusitis and pharyngitis/tonsillitis. (Penicillin is the usual drug of choice in the treatment of *Streptococcus pyogenes* pharyngitis, including the prophylaxis of rheumatic fever).

In sexually transmitted diseases in men and women, **Azithro** is indicated in the treatment of uncomplicated genital infections due to *Chlamydia trachomatis*. It is also indicated in the treatment of chancroid due to *Haemophilus ducreyi*, and uncomplicated genital infection due to non-multiresistant *Neisseria gonorrhoea*, concurrent infection with *Treponema pallidum* should be excluded.

Azithro is indicated, either alone or in combination with rifabutin, for prophylaxis against *Mycobacterium avium-intracellular* complex (MAC) infection, an opportunistic infection prevalent in patients with advance human immunodeficiency virus (HIV).

Azithro is indicated in combination with ethambutol for the treatment of disseminated MAC (DMAC) infection in patients with advanced HIV infection.

3.3 Dosage and Administration

Oral **Azithro** should be administered as a single daily dose.

Capsule should be taken at least 1 hour before or 2 hours after food.

The period of dosing with regard to infection is given below:

In Adults

For the treatment of sexually transmitted disease caused by Chlamydia trachomatis, Haemophilus ducreyi, or susceptible Neisseria gonorrhoea - the dose is 1000 mg as single oral dose.

For prophylaxis against MAC Infections - in patients with the human immunodeficiency virus (HIV), the dose is 1200 mg once per week.

For the treatment of DMAC infections - in patients with advanced HIV infection, the recommended dose is 600 mg once a day.

Azithro should be administered in combination with other antimycobacterial agents, such as ethambutol at the approved dose.

For all other indications in which the oral formulation is administered, the total dosage of 1500 mg should be given as 500 mg daily for 3 days.

In Children

Azithro capsules should only be administered to children weighing more than 45 kg the recommended dose as per adults.

In the Elderly - The same dosage as in adult patients is used in the elderly.

In Patients with Renal Impairment - No dose adjustment is necessary in patients with mild to moderate (GFR 10-80 ml/min) or severe (GFR <10 ml/min) renal impairment.

In Patients with Hepatic Impairment - The same dosage in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

3.4 Contraindication

The use of this product is contraindicated in patients with a history of allergic reaction to azithromycin or any of the macrolide antibiotics.

3.5 Warnings

1. Do not use in hypersensitive patients.

2. This drug may cause liver damage.

3.6 Special Warnings and Precautions for Use

Hypersensitivity

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)], **Azithro** should be discontinued immediately and appropriate treatment should be urgently initiated. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment.

Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued.

Prolongation of the QT Interval

Prolonged cardiac repolarization and QT Interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin (see section 3.9). Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:

- Patients with congenital or documented QT prolongation.
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones.
- Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesemia.
- Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency.
- Elderly patients : elderly patients may be more susceptible to drug associated effects on the QT interval.

Since liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease.

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In patient receiving ergot derivative, ergotism has been precipitated by coadministration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be coadministered. As with antibiotics preparation for signs of superinfection with non-susceptable organisms, including fungi is recommended.

As with any severe renal impairment (GFR <10 ml/min) increase in systemic exposure to azithromycin was observed.

Infantile hypertrophic pyloric stenosis (IHPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and/or irritability with feeding occurs.

3.7 Interaction with Other Medicaments and Other Forms of Interaction

Antacids: The effects of simultaneous administration of antacid with azithromycin no effect on overall bioavailability.

Ceftriaxone: No pharmacokinetic interaction and no significant changes in the QT interval.

Didanosine (Dideoxyinosine): No pharmacokinetic interaction.

Digoxin: Some of the macrolide antibiotics have been reported to impair the microbial metabolism of digoxin in the gut in some patients.

In patients receiving concomitant azithromycin, a related azalide antibiotic, and digoxin the possibility of raised digoxin levels should be borne in mind.

Zidovudine: Single 1000 mg doses and multiple 1200 mg or 600 mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite.

Ergot: Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.

Pharmacokinetic between azithromycin and the following drugs known to undergo significant cytochrome P450 mediated metabolism.

Atorvastatin: Coadministration of atorvastatin (10 mg daily) and azithromycin (500 mg daily) did not alter the plasma concentrations of atorvastatin.

Carbamazepine: No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in subject receiving concomitant azithromycin.

Cimetidine: The effects of a single dose of cimetidine, given 2 hours before azithromycin, on the pharmacokinetics of azithromycin, no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants: Azithromycin did not alter the anticoagulant effect of a single 15-mg dose of warfarin. There have been post-marketing reports of potentiated anticoagulation subsequent to coadministration of azithromycin and coumarin-type oral anticoagulants. Prothrombin time should be carefully monitored when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin: Consequently, caution should be exercised before considering concurrent administration of these drugs. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz: No pharmacokinetic interaction.

Fluconazole: No pharmacokinetic interaction.

Indinavir: No pharmacokinetic interaction.

Methylprednisolone: No pharmacokinetic interaction.

Midazolam: No pharmacokinetic interaction.

Nelfinavir: No pharmacokinetic interaction.

Rifabutin: Coadministration of azithromycin and rifabutin did not effect the serum concentrations of either drug. Neutropenia was observed in subjects receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil: There was no evidence of an effect of azithromycin (500 mg daily for 3 days) on the AUC and C_{max} of sildenafil or its major circulating metabolite.

Terfenadine: There have been rare cases reported where the possibility of such an interaction when azithromycin and terfenadine are coadministered.

Theophylline: No significant effect on any of the pharmacokinetic interaction when azithromycin and theophylline are coadministered.

Triazolam: No significant effect on any of the pharmacokinetic interaction when azithromycin and triazolam are coadministered.

Trimethoprim/sulfamethoxazole: Co-administration of trimethoprim/sulfamethoxazole DS (160 mg/800 mg) for 7 days with azithromycin 1200 mg on the Day 7 had no significant effect on peak concentration, total exposure of urinary excretion of either trimethoprim or sulfamethoxazole.

3.8 Pregnancy and Lactation

Azithro should only be used in pregnant or lactating women where adequate alternatives are not available.

3.9 Effects on Ability to Drive and Use Machines

There is no evidence to suggest that **Azithro** have an effect on a patients ability to drive or operate machinery.

3.10 Undersirable Effects

Azithromycin is well tolerated with a low incidence of side effects.

The following undesirable effects:

Gastrointestinal - Nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps), and flatulence.

Hematopoietic - Transient episodes of mild neutropenia have occasionally been observed.

Liver/Biliary - Abnormal liver function.

Special Senses - Hearing impairment (including hearing loss, deafness and/or tinnitus) has been reported in some patients receiving azithromycin. Many of these have been associated with prolonged use of high doses in investigational studies.

In those cases where follow-up information was available the majority of these events were reversible.

The following undesirable effects in association with DMAC prophylaxis and treatment:

The most frequent adverse reactions in HIV-infected patients receiving azithromycin for prophylaxis for DMAC were diarrhea, abdominal pain, nausea, loose stools, flatulence, vomiting, dyspepsia, rash, pruritus and arthralgia.

When azithromycin 600 mg is given daily for the treatment of DMAC infection for prolonged periods, the most frequently reported treatment-related side effects are abdominal pain, nausea, vomiting diarrhea, flatulence, headache, abnormal vision and hearing impairment.

The following additional undesirable effects:

Body as a Whole - Asthenia, although a causal relationship has not been established; fatigue, malaise, moniliasis, and anaphylaxis (rarely fatal).

Post-marketing experience:

Cardiac Disorders: Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongation and torsades de pointes (see Special Warnings and Precautions for Use).

Central & Peripheral Nervous System - Dizziness/vertigo, convulsions (as seen with other macrolides), headache, hyperactivity, paresthesia, somnolence, and syncope.

Gastrointestinal Disorders - Anorexia, dyspepsia, constipation, pseudomembranous colitis, pancreatitis, rare reports of tongue discoloration, vomiting/diarrhea (rarely resulting in dehydration), infantile hypertrophic pyloric stenosis.

Genitourinary - Interstitial nephritis and acute renal failure.

Hematopoietic - Thrombocytopenia.

Liver/Biliary - Hepatitis and cholestatic jaundice.

Musculoskeletal - Arthralgia

Psychiatric - Aggressive reaction, nervousness, agitation and anxiety.

Reproductive - Vaginitis.

Skin and Subcutaneous Tissue Disorders: Allergic reactions including pruritus, rash, photosensitivity, edema, urticaria and angioedema. Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

Special Senses - There have been rare reports of taste perversion.

3.11 Overdosage

Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. In the event of overdosage, general symptomatic and supportive measures are indicated as required.

4. STORAGE CONDITION: Store below 30°C.

5. PRODUCT PRESENTATION

Azithro (Capsule) White powder in capsule #0, imprinted with M&H in black on white cap and body, in the box of 6 capsules

Manufactured by: M & H MANUFACTURING CO., LTD.

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Product Registration Number : MAL10100178AZ

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